

DeepSeek-V2

Efficient MoE Language Model with
Multi-Head Latent Attention

236B

Total Parameters

21B

Activated per Token

128K

Context Length

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

Larger dense models yield better intelligence but incur prohibitive training costs and slow inference due to KV cache bottlenecks.



Performance Trap

Scaling dense models improves benchmarks but grows compute at $O(N)$ for every activated parameter.



Training Cost

Pretraining a 70B dense model costs tens of millions of dollars, limiting who can afford to train.



KV Cache Bottleneck

Standard MHA caches full key-value pairs per head per token, making long-context inference memory-bound.

→ *MoE + efficient attention are needed to break the cost-performance trade-off*

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block

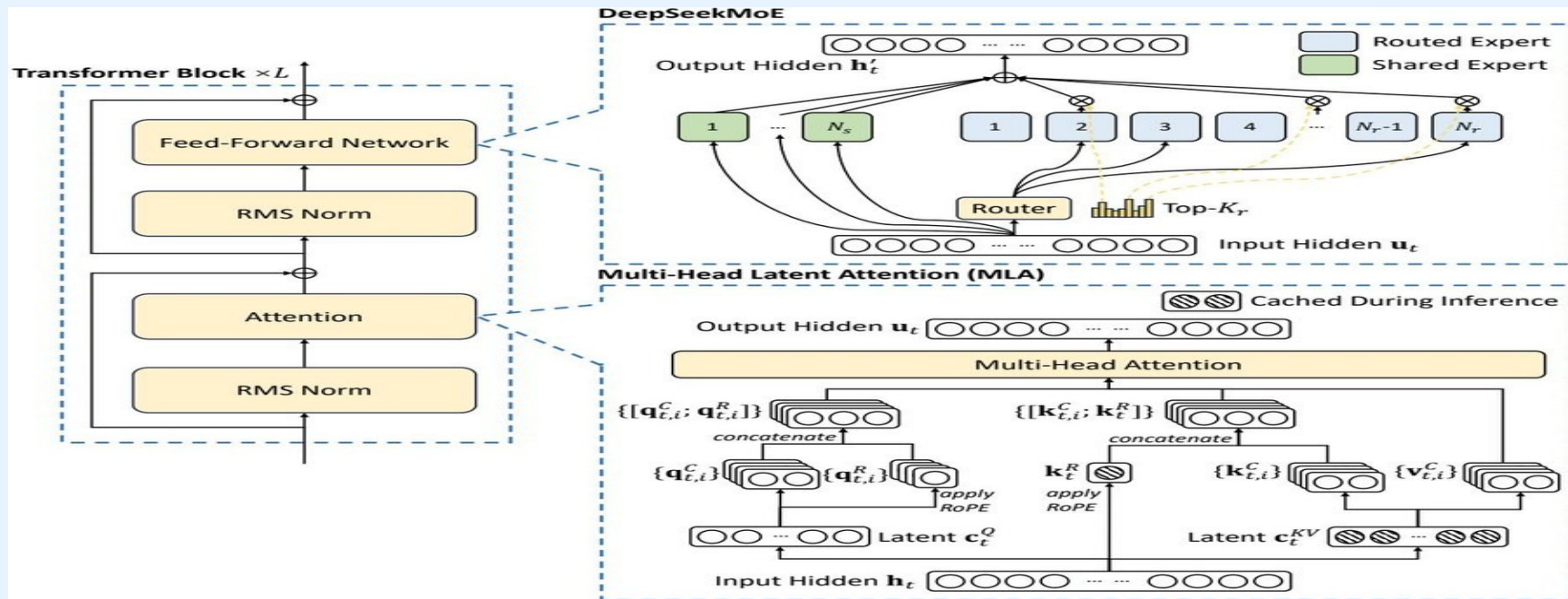
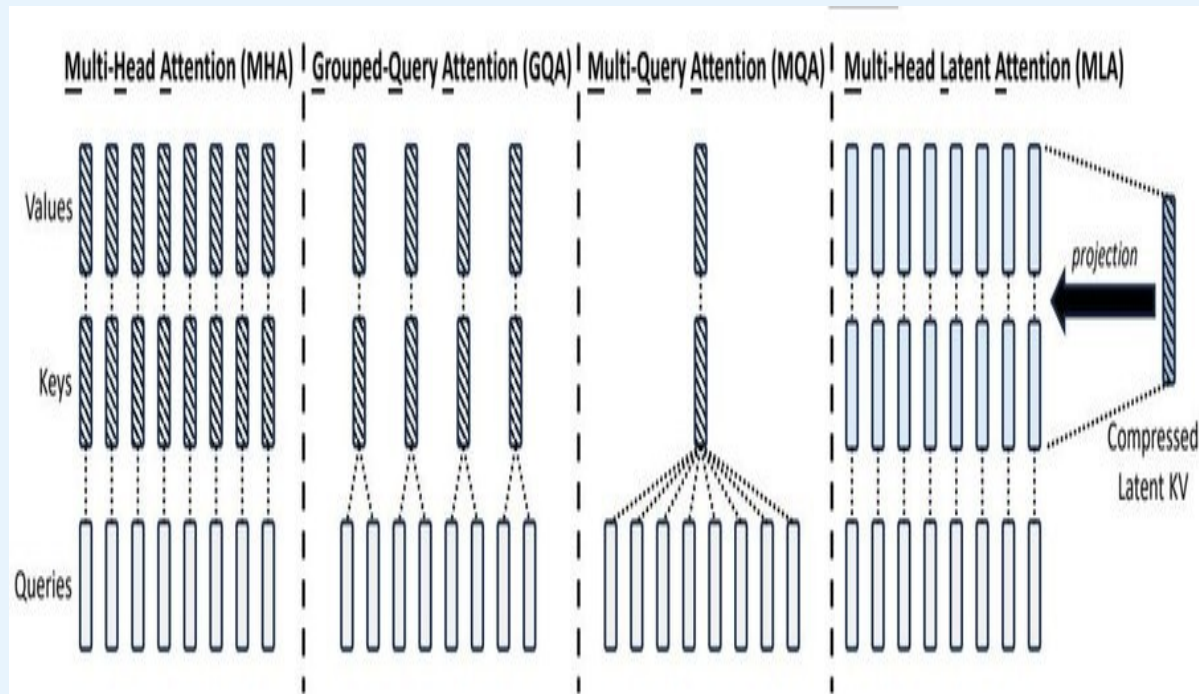


Figure 1: DeepSeek-V2 block with Multi-Head Latent Attention (MLA) and DeepSeekMoE replacing standard attention and dense FFN.

MLA replaces standard MHA: low-rank KV compression
→ 93.3% cache reduction

DeepSeekMoE replaces dense FFN: shared + routed experts, top-K routing

Multi-Head Latent Attention Compresses KV Cache by 93.3%



MHA (Baseline)

Full K,V per head per token. High memory, best quality.

GQA / MQA

Shared K,V across heads. Less cache but degraded quality.

MLA (Ours)

Low-rank KV joint compression + decoupled RoPE. Outperforms MHA with 93.3% less cache.

Key insight: Compressing K,V into a low-rank latent vector c_t^{KV} and reconstructing on-the-fly via learned projections yields better performance than MHA while requiring only 6.7% of the cache.

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Expert Architecture

Shared Experts (N_s)

Always activated. Capture common knowledge shared across all tokens. Reduce redundancy in routed experts.

Routed Experts (N_r)

Sparsely activated via top-K router. Fine-grained segmentation increases specialization potential.

Training Efficiency Mechanisms



Fine-grained expert segmentation increases specialization granularity



Device-limited routing: each token sent to experts on $\leq M$ devices $\rightarrow$ reduces cross-node communication



Auxiliary load-balancing losses prevent expert collapse



Token-dropping strategy during training prevents expert overflow (inference unchanged)

Result: 236B total parameters — activating only 21B per token — trained at 42.5% lower cost than DeepSeek 67B dense model

42.5% Training Cost Reduction & 5.76× Throughput vs. DeepSeek 67B

DeepSeek-V2 relative efficiency gains over its dense predecessor



42.5%

**Training Cost
Reduction**

vs. DeepSeek 67B



93.3%

**KV Cache
Reduction**

vs. DeepSeek 67B



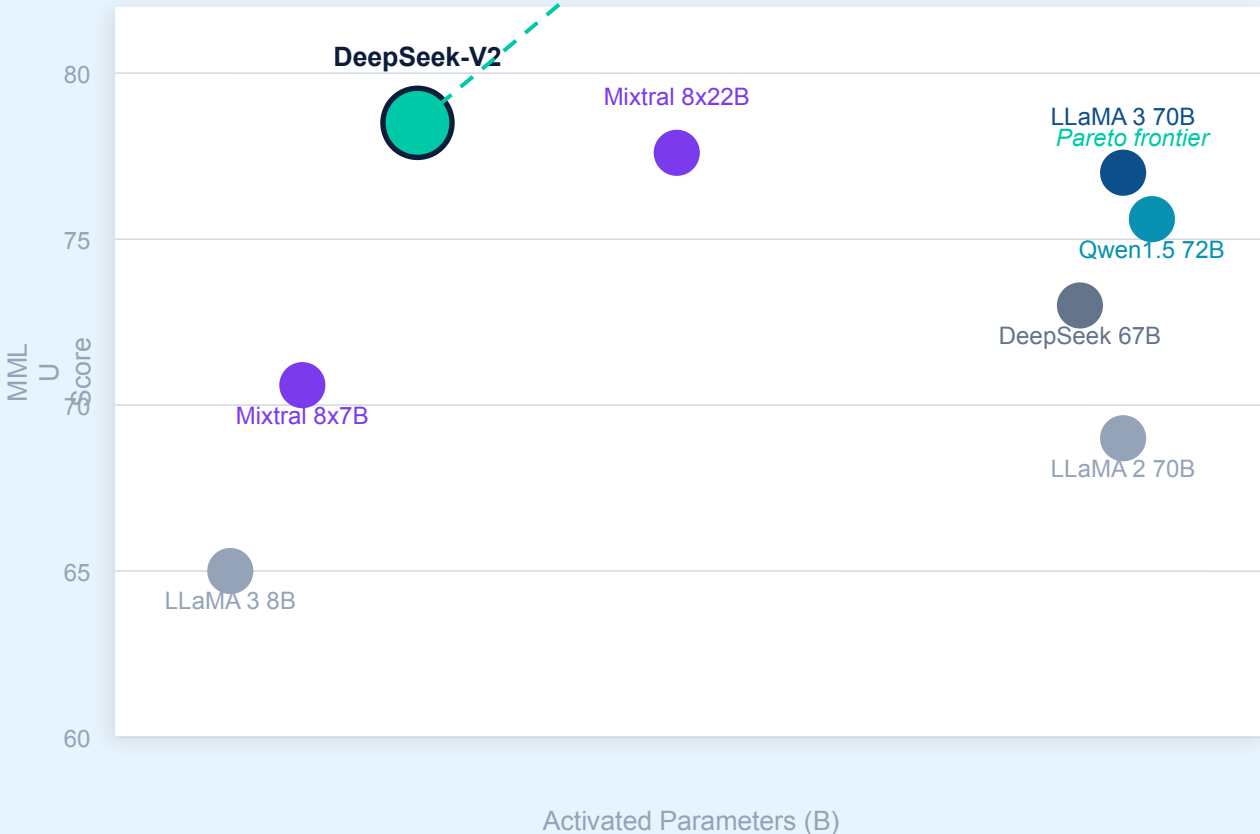
5.76×

**Generation
Throughput**

max vs. DeepSeek 67B

No absolute FLOPs or absolute cache values fabricated — all metrics are relative improvements over DeepSeek 67B as reported.

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activated Parameters



MMLU vs. Activated Params

Model	Act. B	MMLU
DeepSeek-V2	21	78.5
LLaMA 3 70B	70	77.0
Mixtral 8x22B	39	77.6
Qwen1.5 72B	72	75.6
DeepSeek 67B	67	73.0
LLaMA 2 70B	70	69.0
Mixtral 8x7B	13	70.6
LLaMA 3 8B	8	65.0

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

Pretraining
8.1T tokens

SFT
1.5M sessions

GRPO RL
Group Relative Policy Opt.

AlpacaEval 2.0

Length-controlled win rate

38.9%

Outperforms most open-source

MT-Bench

Overall score

8.97

Competitive with closed models

AlignBench

Chinese overall

7.91

Best open-source in Chinese

On AlignBench (Chinese), DeepSeek-V2 Chat (RL) outperforms all open-source and most closed-source models.

Key Takeaways

Architectural Innovation Unlocks Efficient Scaling

- ✓ MLA achieves superior performance to MHA while reducing KV cache by 93.3% — a strictly better alternative to GQA/MQA for inference efficiency.
- ✓ DeepSeekMoE with fine-grained experts and shared expert isolation enables training a 236B-parameter model at 42.5% lower cost than a 67B dense model.
- ✓ Only 21B activated parameters are needed to match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU.
- ✓ 5.76× generation throughput improvement over DeepSeek 67B makes DeepSeek-V2 practical for large-scale deployment.
- ✓ DeepSeek-V2 Chat (RL) is the strongest open-source chat model in Chinese and competitive with closed-source systems in English open-ended evaluation.